SLO & Error Budget – Two-Tier Web Service

1. Introduction:

Service Level Indicators (SLIs), Service Level Objectives (SLOs), and error budgets help measure the reliability and performance of a web service. They help the team monitor service health, respond to incidents, and balance reliability with development activities.

This document defines three service scenarios: Lean, Balanced, and Resilient. The targets are based on the provided SLO and cost-envelope worksheet.

2. Service Level Indicators (SLIs):

SLIs are measurable values used to evaluate the service's reliability and performance.

SLI	Description	Measurement
Availability	Measures whether the service successfully handles valid requests	Successful requests ÷ Total valid requests × 100
Latency	Measures how quickly the service responds	95th percentile (p95) response time
Error Rate	Measures the percentage of failed requests	Failed requests ÷ Total requests × 100
Database Health	Measures database responsiveness and connection health	Query latency and connection errors

3. Service Level Objectives (SLOs):

SLOs define the target availability and response time for each operating scenario.

Metric	Lean	Balanced	Resilient
Monthly Budget	₹2,500	₹8,000	₹25,000
Availability Target	99%	99.5%	99.9%
p95 Latency Target	800 ms	500 ms	300 ms
Recovery Point Objective (RPO)	24 hours	1 hour	15 minutes
Recovery Time Objective (RTO)	4 hours	1 hour	30 minutes
Expected Users	500	5,000	25,000

These are target values from the planning worksheet. Actual service performance must be measured using monitoring data.

4. Error Budget Calculation:

An error budget is the amount of unreliability permitted while still meeting the availability SLO.

Formula:

Error Budget = 100% − Availability SLO

For a 30-day month containing 43,200 minutes:

Allowed Downtime = 43,200 × (1 − Availability Target)

Scenario	Availability SLO	Error Budget	Approximate Monthly Downtime
Lean	99%	1%	432 minutes (7 hours 12 minutes)

Balanced	99.5%	0.5%	216 minutes (3 hours 36 minutes)
Resilient	99.9%	0.1%	43.2 minutes

These downtime figures assume a 30-day month. The actual error budget should be calculated using the agreed measurement window and availability definition.

5. Alerting and Paging Thresholds:

The following are proposed operational alert thresholds. They are recommendations for monitoring and are not separate targets specified in the original worksheet.

Severity	Condition	Action
Critical / P1	Service unavailable or error budget being consumed rapidly	Page the on-call engineer immediately
High / P2	Availability falls below the selected scenario target	Investigate and notify the responsible team
Warning	p95 latency exceeds the selected scenario target for 10 minutes	Investigate application and database performance
Informational	CPU or memory utilization remains above 80% for 15 minutes	Review resource usage and scaling requirements

Latency alert targets should match the selected scenario: 800 ms for Lean, 500 ms for Balanced, and 300 ms for Resilient.

6. Incident Response:

1. Detect the issue through monitoring or an alert.
2. Assess the severity and impact on users.
3. Identify whether the application, database, network, or deployment caused the issue.
4. Apply mitigation, such as rollback, restart, scaling, or safe recovery.
5. Verify that service performance has returned to acceptable levels.
6. Record the incident, root cause, downtime, and preventive actions.

7. Recovery Objectives:

Recovery objectives define the expected recovery time and acceptable data-loss window after a major disruption.

Scenario	Recovery Time Objective (RTO)	Recovery Point Objective (RPO)
Lean	4 hours	24 hours
Balanced	1 hour	1 hour
Resilient	30 minutes	15 minutes

RTO is the target time to restore service. RPO is the maximum targeted data-loss window.

Achieving these targets depends on backup frequency, database recovery capabilities, infrastructure, and tested recovery procedures. They should be validated through recovery tests.

8. Error Budget Policy:

- **Lean:** Prioritize cost-effective reliability measures and monitor service failures against the 99% availability target.
- **Balanced:** Use regular monitoring, automated backups, and incident response procedures to support the 99.5% availability target.
- **Resilient:** Use stronger recovery capabilities, frequent backups, and rapid incident response to support the 99.9% availability target.
- If error-budget consumption increases rapidly, investigate the root cause and prioritize reliability improvements.
- If the error budget is exhausted, consider pausing non-essential releases until service reliability improves.

The appropriate measures depend on the budget, architecture, and business requirements of the selected scenario.

9. Review and Reporting:

Review availability, p95 latency, error rates, incidents, and error-budget consumption regularly. Compare measured performance with the selected SLO targets and update the plan when workload or business requirements change.

10. Conclusion:

SLIs measure service performance, SLOs define reliability targets, and error budgets indicate how much unreliability is acceptable. The Lean, Balanced, and Resilient scenarios provide different levels of service reliability and recovery capability to support different budgets and user requirements.